PROJECT REPORT: Used Car Price Forecasting in Seattle, Washington

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INTRODUCTION

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Research Problem:

Price prediction has always been a challenging task as there are several factors involved that tend to cause variation in prices. The used car market in Seattle, Washington, is influenced by a unique combination of factors, including urban demand, environmental policies favoring electric vehicles (EVs), and fluctuating economic conditions.

Accurately forecasting car prices in this region is crucial for buyers, dealerships, and financial institutions to make informed decisions. However, price variations depend on multiple factors such as brand, model, year, mileage, engine size, fuel type, transmission type, ownership history, and vehicle configuration.

This research aims to develop a predictive model for used car prices in Seattle, considering these attributes to enhance pricing accuracy and market understanding.

Why this research problem is important

Our main objective is to identify how effectively machine learning models, namely multiple linear regression model and Decision trees can be incorporated to predict used car prices based on the “car price dataset” to predict the price of the used car prices in Seattle, Washington and compare the accuracy levels and reliability of the models.

To acquire the minimal error rate and residual sum of squares, the selection of attributes plays a vital role, as the models might be biased on the test data set. The dataset primarily consists of attributes that are categorical and quantitative including brand, model, year, mileage, fuel type which have a direct impact on the car price depreciation over time.

Upon review of research papers done previously by other scholars, we found evidence that despite their results yielding applicable models, there is a need to further improve and develop a deeper understanding of used car price prediction models.

For example, one paper, Pillai (2022) uses an Artificial Neural Network (ANN), trained on data scraped from Kaggle and other car marketplaces. The model in this case was chosen because it handles complex, non-linear relationships well and can learn intricate patterns in high-dimensional data (e.g., brand, model, mileage, and year). However, one of the drawbacks of ANN models is that they are complex and difficult to fully interpret, especially in explaining the model architecture and training process, this is something our research will aim to improve upon by keeping our results interpretability high utilizing Clear visuals and tables so as to improve understanding of our model.

Another paper, Rane et al.(2021) used regression algorithms—specifically: Linear Regression, Ridge Regression and Lasso Regression. These methods are appropriate because the target variable (car price) is continuous. These algorithms are designed to model relationships between input features (e.g., age, mileage, and brand) and a continuous outcome, making them well-suited for price prediction tasks. This paper also compared model performance to choose the best one, which is a good practice in machine learning workflows which we will also adopt.

However, while that paper mentions model accuracy and usability, there is no clear presentation of accuracy metrics (e.g., RMSE, R^2) or a dataset description, which weakens the rigor of the conclusion. For full confidence, our approach will provide more empirical results by following a more defined Data Collection methodology, utilizing a real-world dataset from Kaggle, and feature EDA (Exploratory Data Analysis) whereby we will visualize relationships, detect outliers, handle missing values as well.

How does it relate to the STAT 4600 class?

Our research problem—predicting used car prices in Seattle, Washington—is an excellent and practical application of data analytics, and here's why it relates to our STAT 4600 class:

1. Real-World Relevance and Practical Impact

Understanding how car prices fluctuate based on various features (like mileage, make, year, transmission, etc.) is not just an academic exercise. In Seattle—a city with unique environmental policies, a growing EV market, and shifting economic conditions—this type of model can help:

- ✓ Buyers avoid overpaying.
- ✓ Dealers set competitive prices.
- ✓ Lenders and insurers assess asset value more precisely.

By conducting this research, we are addressing a real-world problem with direct economic implications, making our findings useful outside the classroom.

2. Application of Statistical Techniques

This project provides a valuable opportunity to practically apply the core concepts covered in our statistics class. Through hands-on experience, we are able to implement techniques such as multiple linear regression, classification and regression trees, correlation analysis, and model diagnostics with a focus on validating assumptions.

Rather than merely understanding these concepts in theory, we are learning how to build, test, and refine predictive models using real-world data, enhancing both our analytical skills and our ability to draw meaningful insights from statistical methods.

We are not just learning theory—we are learning how to build, test, and refine predictive models using real data.

3. Hands-On Learning with R Studio

R Studio is a powerful environment for data analysis and visualization, offering a comprehensive set of tools to work with data effectively. Through this research, we will gain hands-on experience with various aspects of data manipulation and modeling. This includes data cleaning and wrangling using packages like dplyr and tidyr, which help prepare the data for analysis.

We will also explore data visualization using ggplot2 to create insightful graphs and plots. Additionally, we will build predictive models with functions such as lm() for linear regression and rpart for decision trees.

Finally, we will evaluate model accuracy using metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 to assess how well our models perform. These are marketable skills in data science, analytics, and research roles.

Type/source/content of data used in the project

Type of Data: The dataset used in this project is a collection of car price prediction variable information from Seattle, Washington, spanning car models from 2000 -2023.

Source of Data: The dataset is sourced from Kaggle.com, specifically from the dataset titled "Car price dataset" which can be accessed here: https://www.kaggle.com/datasets/asinow/car-price-dataset.

About the dataset: The dataset comprises various variables for the top 10 American car producers, listed below;

1. **Brand**: Different types of Brands of cars.
2. **Model**: particular vehicle name.
3. **Year**: Year of manufacturer.
4. **Engine Size**: Vehicle Engine fuel capacity.
5. **Fuel Type**: Particular vehicle fuel.
6. **Transmission**: Vehicle drive train.
7. **Mileage**: Total vehicle distance driven.
8. **Doors**: Number of doors on vehicle.
9. **Owner Count**: Number of vehicle owners from when new.
10. **Price**: Listed price of the car.

Modeling Methods

Analysis Using Decision Trees

1. Objective

The aim is to build a Decision Tree regression model that predicts the price of used cars in Seattle based on features such as year, mileage, brand, engine size, fuel type, transmission, and other relevant attributes in the dataset, thus determining which variables have the greatest impact predicting the price of used cars.

2. Why Decision Trees?

Decision Trees are a powerful and intuitive machine learning method, particularly well-suited for this case due to several key advantages. First, they offer a visually interpretable model, making it easier to understand and communicate insights. They also handle datasets containing a mix of numerical and categorical variables effectively.

One of the major strengths of decision trees is their ability to automatically capture complex interactions between variables without the need for manual specification. Unlike linear regression, decision trees do not require any assumptions about the underlying data distribution, making them more flexible.

Additionally, they are robust to missing values and outliers, which enhances their reliability in real-world scenarios.

3. Steps in the Analysis

Data preparation involves several crucial steps to ensure the dataset is ready for analysis. First, the dataset is imported and cleaned using R, which includes checking for and addressing any inconsistencies.

Categorical variables are then converted into factors to facilitate appropriate modeling. Irrelevant or redundant features are removed to streamline the dataset and improve model performance.

Finally, any missing values are handled, ensuring the data is complete and suitable for further analysis.

b. Splitting the Data

We then Split the dataset into training and testing sets, namely by a 70/30 split.

c. Building the Decision Tree Model

We then used the rpart package to build a regression tree, and examined its complexity parameter table (*Table 2.)* to establish whether our decision tree requires pruning or not.

In this case, our tree was found to not need any pruning as we able to establish through plotting the complexity parameter (*Fig 1.)* that our tree has the lowest cross validated error and thus our complexity parameter is optimal.

d. Visualizing the Tree

Interpreting the visual model (*Fig 2.*) is how we are able to make decisions, by observing that—the first split is by year, then by mileage and finally engine size. We are thus able to identify the most important variables to our model accurately and establish their different weights on the outcome of price prediction.

e. Model Evaluation

We are now able to Predict car prices on the test data through use of predict () function, based on the CART model we created.

We are able assess the **R-squared** to understand how well the model explains the variance. We find an R-squared of 0.7 ~ 70%, meaning our CART model explains 70% of the variance in predicting price.

4. Interpretation of the Tree

The top split, or root node, in a decision tree reveals the most important feature for determining the target variable which in this case, the year of the car's manufacture, which plays a significant role in predicting the price.

As we move down the tree, each branch represents a decision rule, illustrating how different variables influence price predictions. These decision rules take into account the number of cars in the sample data that each variable affects, as well as the percentage of those cars that fall under each rule.

This structure allows for a clear understanding of how each feature contributes to the final price prediction.

5. Conclusion

1. **Key Drivers of Price**

   The tree's variable-importance rankings (*Table 1*.) highlighted that Year, Mileage, and engine size were the most influential factors in determining price, followed by Model, Brand and Fuel type. These findings align with industry expectations: newer, lower-mileage vehicles command higher prices, while fuel economy and premium brands with specific model lines introduce further variation.

2. **Model Performance**

   On the held-out test set, the CART model achieved an MSE of 2932321 and an R^2 of 0.7, indicating that approximately 70% of price variability is captured by our tree. While these metrics show reasonably good predictive power, there remains room to reduce error and boost explained variance with other types of model testing i.e. linear regression.

3. **Overfitting and Pruning**

   the complexity-parameter (cp) table (*Table 2*.) and CP-error plot (*Fig 3*.) revealed a deeper tree with a low cross-validated error, removing the need to prune away branches that contribute minimal predictive gain.

Overall, the CART regression tree provides a transparent, intuitive model that surfaces the most important determinants of car pricing.

Analysis Using Multiple Linear Regression

1. Objective

The goal is to develop a predictive model for estimating the price of used cars in Seattle using multiple linear regression (MLR)**.** MLR allows us to examine how multiple independent variables simultaneously affect a single continuous dependent variable—car price in this case.

2. Why Multiple Linear Regression?

Multiple linear regression is well-suited for this problem because the response variable, car price, is continuous, making it appropriate for regression analysis. Additionally, there are multiple predictors, both numerical such as mileage, year, and engine size—and categorical—such as fuel type, transmission, and brand.

This approach allows for the understanding of how each factor contributes to the car price while controlling for the influence of other variables. By analyzing these relationships, multiple linear regression helps quantify the individual effect of each predictor on the outcome.

3. Steps in the Analysis

a. Data Preparation

The process begins by importing the dataset into R using the read.csv() function. Next, the data is cleaned by handling missing values, converting categorical variables to factors, and removing any duplicates to ensure the dataset is accurate and consistent.

Once the data is prepared, Exploratory Data Analysis (EDA) is performed using functions like summary() to get an overview of the dataset, cor() to examine correlations between numerical variables, and ggplot2 for visualizing distributions and relationships within the data.

This step provides valuable insights and helps identify patterns or anomalies in the dataset.

b. Model Building

Using the lm() function in R:

We used the multiple regression models to investigate the effects of different variables (Fig 3.), namely brand, model, mileage, engine size, transmission, and fuel type, on the price of used cars in Seattle, Washington.

Using the Adjusted R of the multiple linear regressions, the variation in the price is 99.9% explained by variables within the model. This explains an extremely good fit. The residual standard error is 85.76, which is small, and this also supports that our model`s prediction is close to the actual data points.

c. Model evaluation

To validate assumptions of linear regression:

The significant variables were found to be year ($p < 0.001$), brand ($p = 0.014$), engine size, Fuel Type (Electric/Hybrid/Petrol), Transmission (Manual/Semi-Automatic), and Mileage.

4. Interpretation of Results

 Year of Manufacture: Each additional model year increases the vehicle's price by **$298.60**, indicating a steady, positive linear trend.

 Engine **Size**: Every one-liter increase in engine displacement is associated with a **$992.70** rise in price, highlighting the premium on larger engines.

 Transmission Type:

- Manual transmissions are linked to a **$1,490** decrease in price.
- Semi-automatic transmissions are linked to a **$1,492** decrease in price.

 Mileage: An increase of 10,000 miles driven corresponds to a **$199** reduction in resale value, reflecting the expected negative mileage impact.

 Fuel Type (vs. reference category):

Fuel Type (Electric/Hybrid/Petrol) has a linear and positive relationship based on the category when electric is 1 and hybrid and petrol cars are 0, and otherwise. The observations are as stated below:

- When electric id coded 1, Electric vehicles command **$1,988** more.
- When the hybrid is coded 1, Hybrids command **$9,904** more.
- When petrol is coded 1, Petrol vehicles command **$1,240** more.

 Brand Premiums (vs. reference brand):

- **Honda** models carry a **$9,290** premium.
- **KIA** models carry a **$9,740** premium.

Newer vehicles, premium brands, larger engines, and environmentally friendly fuel types (electric/hybrid) increase value corroborating previous research projects done, i.e. Hollenbacher(2015), and James *et al*(2012).

These researchers found that every added mile decrease car prices by 5 – 10%. Manual transmission and high mileage lower the prices. Such findings become very valuable in the Seattle economy, which is friendlier to the environment and is therefore characterized by a higher demand for hybrid and electric cars that thus command premium prices.

Results

In this section, we present the key findings from our analysis into the factors driving used-car prices in Seattle. To support a robust and nuanced exploration, we employed both classification and regression tree analysis alongside traditional linear regression.

This dual-method approach allowed us to accurately quantify the relative importance and directional effects of each variable on market valuation, setting the stage for a detailed discussion of how year of manufacture, engine size, mileage, transmission type, fuel technology, and brand prestige influence buyer willingness to pay.

- **Year of Manufacture**: Exhibits the strongest impact on predicted price—newer vehicles command significantly higher values.
- **Engine Size**: Larger engines are associated with increased prices, reflecting buyer willingness to pay a premium for performance.
- **Mileage**: Lower-mileage cars consistently achieve higher prices, underscoring the importance of vehicle condition.
- **Transmission Type**: Manual and semi-automatic transmissions carry a price penalty, indicating Seattle buyers prefer automatic models.
- **Fuel Type**: Electric and hybrid vehicles enjoy a positive price premium over traditional petrol and diesel models, highlighting a strong preference for fuel efficiency.
- **Brand and Model**: Premium marques—such as KIA and Honda (Japan)—and their high-end model lines introduce additional price variation, with certain models commanding top dollar.

Business Insights and recommendations

1. **Newer Vehicles Drive Value**

   The year of manufacture has the greatest influence on price. Newer vehicles consistently sell for more, indicating consumer preference for recent models and up-to-date features. Retailers should Focus Inventory on Recent Models**;** Stock newer vehicles (within 3–5 years of manufacture) to maximize profitability and align with buyer preferences.

2. **Performance Matters**

   Larger engine sizes correlate with higher prices, suggesting that a segment of buyers is willing to pay more for better performance or perceived power.

3. **Condition is Crucial**

   Lower mileage significantly boosts resale value, reinforcing that buyers strongly associate mileage with reliability and longevity. Retailers should promote vehicles with lower mileage prominently in marketing materials and pricing strategies.

4. **Transmission Preferences are Regional**

   Manual and semi-automatic transmissions are penalized in pricing, indicating that Seattle buyers clearly prefer automatic vehicles. Retailers should Prioritize automatic models in inventory sourcing and reduce reliance on manual/semi-automatic listings in the Seattle market.

5. **Fuel Efficiency is a Selling Point**

   Electric and hybrid vehicles fetch higher prices, showing a clear shift toward eco-conscious and fuel-efficient transportation. Retailers should capitalize on Green Vehicle Demand and Expand offerings of electric and hybrid vehicles. Consider partnerships with EV dealerships or suppliers to meet demand.

6. **Brand Premium Exists**

   Certain brands, particularly Japanese marques like Honda and KIA, have a strong price.

   Retailers should Source High-Value Brands Strategically and Focus acquisition efforts on

high-performing brands like Honda and KIA, especially their well-known, high-end models—pointing to brand trust and perceived reliability.

7. **Bundle Performance and Efficiency**

   Retailers should look for models that combine larger engine options with hybrid or electric technology to meet the dual demand for power and efficiency.

APPENDIX

```
#Variable importance from tree
importance_vals <- cart_model$variable.importance
importance_df   <- data.frame(
  Feature    = names(importance_vals),
  Importance = as.numeric(importance_vals)
)
print(importance_df)
```

```
##        Feature  Importance
## 1         Year 25597058942
## 2      Mileage 16422233815
## 3  Engine_Size  7200963347
## 4        Model  2126946200
## 5        Brand   790886055
## 6    Fuel_Type   451722989
## 7 Transmission   170542590
## 8  Owner_Count    70939317
## 9        Doors    61104290
```

Table 1.

```
#Examine complexity parameter table and plot
printcp(cart_model)
```

```
##

## Regression tree:

## rpart(formula = Price ~ ., data = train_data, method = "anova",

##     control = rpart.control(cp = 0.01))

##

## Variables actually used in tree construction:

## [1] Engine_Size Mileage     Year

##

## Root node error: 6.7715e+10/7001 = 9672215

##

## n= 7001
```

```
##
##          CP nsplit rel error  xerror      xstd
## 1  0.325660      0   1.00000 1.00018 0.0147610
## 2  0.129783      1   0.67434 0.67457 0.0100351
## 3  0.107317      2   0.54456 0.54509 0.0086905
## 4  0.029440      3   0.43724 0.44169 0.0067928
## 5  0.027673      4   0.40780 0.42023 0.0064857
## 6  0.021329      5   0.38013 0.38812 0.0059356
## 7  0.020587      6   0.35880 0.36920 0.0056562
## 8  0.014035      7   0.33821 0.35029 0.0053080
## 9  0.013999      8   0.32418 0.33790 0.0051773
## 10 0.012426      9   0.31018 0.32679 0.0050327
## 11 0.010074     10   0.29775 0.31093 0.0048088
## 12 0.010000     11   0.28768 0.29524 0.0045340
```

Table 2.

CP PLOT

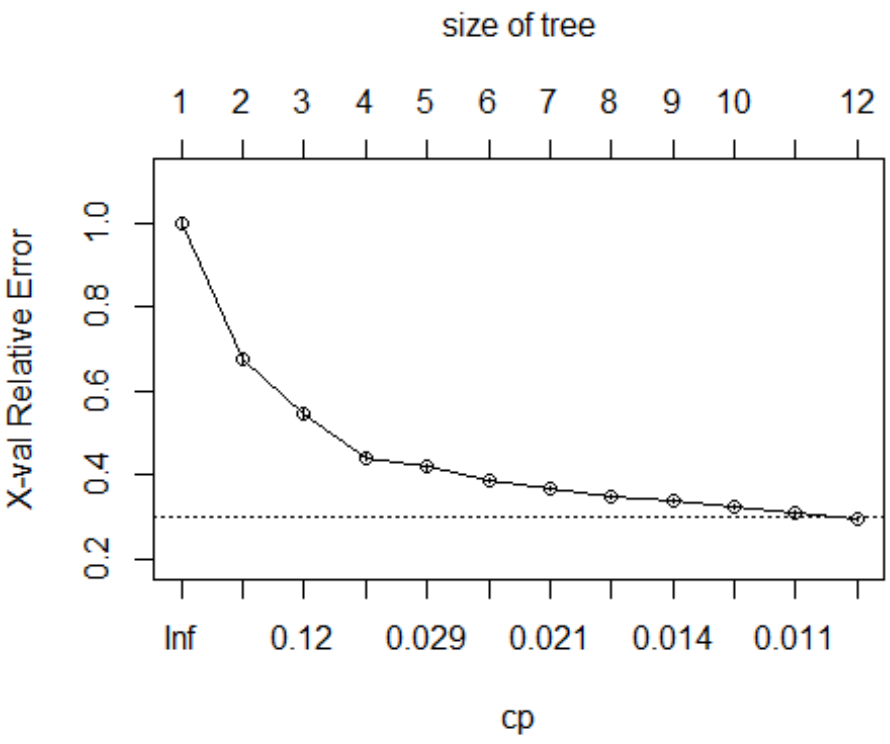

Figure 1.

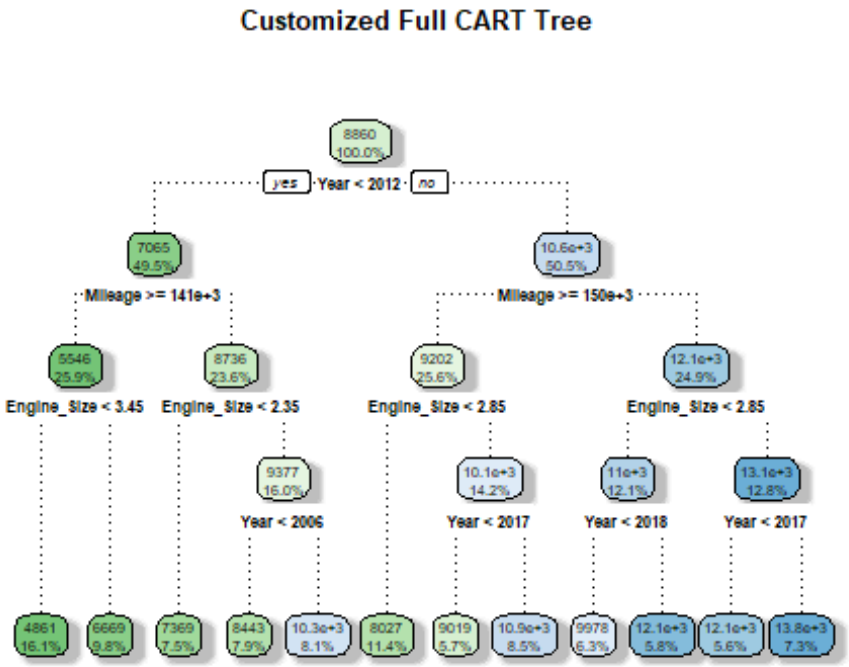
Customized Full CART Tree
8860
100.0%
yes
Year < 2012
no
7065
49.5%
10.6e+3
50.5%
Mileage >= 141e+3
Mileage >= 150e+3
5546
25.9%
8736
23.6%
9202
25.6%
12.1e+3
24.9%
Engine_Size < 3.45
Engine_Size < 2.35
Engine_Size < 2.85
Engine_Size < 2.85
9377
16.0%
10.1e+3
14.2%
11e+3
12.1%
13.1e+3
12.8%
Year < 2006
Year < 2017
Year < 2018
Year < 2017
4861
16.1%
6669
9.8%
7369
7.5%
8443
7.9%
10.3e+3
8.1%
8027
11.4%
9019
5.7%
10.9e+3
8.5%
9978
6.3%
12.1e+3
5.8%
12.1e+3
5.6%
13.8e+3
7.3%

Figure 2.

```
Call:
lm(formula = Price ~ . - Model - Doors - Owner_Count, data = car_data)

Residuals:
    Min      1Q  Median      3Q     Max 
 -55.82  -18.75   -6.37    6.34 2442.45 

Coefficients:
                            Estimate Std. Error   t value Pr(>|t|)    
(Intercept)               -5.916e+05  2.504e+02 -2362.858   <2e-16 ***
BrandBMW                   2.545e+00  3.804e+00     0.669   0.5035    
BrandChevrolet             2.028e+00  3.798e+00     0.534   0.5933    
BrandFord                  3.797e+00  3.756e+00     1.011   0.3121    
BrandHonda                 9.290e+00  3.793e+00     2.449   0.0143 *  
BrandHyundai               1.502e+00  3.806e+00     0.395   0.6930    
BrandKia                   9.740e+00  3.824e+00     2.547   0.0109 *  
BrandMercedes              6.739e+00  3.861e+00     1.745   0.0809 .  
BrandToyota                2.531e+00  3.832e+00     0.661   0.5089    
BrandVolkswagen            1.740e+00  3.782e+00     0.460   0.6454    
Year                       2.986e+02  1.244e-01  2400.113   <2e-16 ***
Engine_Size                9.927e+02  7.470e-01  1328.994   <2e-16 ***
Fuel_TypeElectric          1.988e+03  2.394e+00   830.269   <2e-16 ***
Fuel_TypeHybrid            9.904e+02  2.436e+00   406.580   <2e-16 ***
Fuel_TypePetrol            1.240e+00  2.446e+00     0.507   0.6122    
TransmissionManual        -1.490e+03  2.098e+00  -710.376   <2e-16 ***
TransmissionSemi-Automatic -1.492e+03  2.108e+00  -707.625   <2e-16 ***
Mileage                   -1.989e-02  9.944e-06 -2000.208   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 85.76 on 9982 degrees of freedom
Multiple R-squared:  0.9992,	Adjusted R-squared:  0.9992 
F-statistic: 7.742e+05 on 17 and 9982 DF,  p-value: < 2.2e-16
```

Figure 3.

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